Trained Small, Scored Fairly: A Sub-Million-Parameter Graph Model Against Published Specialists and Zero-Shot Foundation Models on Physical Network Forecasting

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ABSTRACT

Time-series foundation models now forecast any series zero-shot, while graph specialists are still trained per network, and the two are rarely compared under one protocol. We compare them on physical sensor networks with a fixed, open benchmark: six traffic networks (PEMS03/04/07/08, METR-LA, PEMS-BAY) and two monsoon rainfall regions, four tracks (standard context, open context, 30% dead sensors, zero-shot transfer), one scorer for point error, pinball loss and interval score, saved forecasts and paired Diebold–Mariano and block-bootstrap tests. We introduce PWM-Dyn, a 0.3–0.4M-parameter graph-diffusion model with explicit missing-data inputs, quantile outputs and split-conformal calibration, and compare it with STID, Graph WaveNet and STAEformer trained with their official code on identical windows under a fixed compute budget, and with zero-shot TimesFM 2.5 and 3.0. Configurations were chosen on validation and the reported models use fresh seeds; four networks were never used during development. Under a 12-step context a three-seed PWM-Dyn ensemble has the lowest MAE on five of six traffic networks (0.3–4.2% below the best trained opponent on the four new ones), is not significantly different from STID on three (PEMS08, PEMS04, PEMS07), and trails a budget-capped STAEformer on PEMS08. Against TimesFM 3.0 it has lower MAE and interval score on seven of eight tasks (five of six traffic networks; the MAE difference is significant on four; rainfall entries from the previous model version) and loses on PEMS03. With 30% of sensors dark it beats per-sensor climatology on seven of eight tasks, where specialists without a missing-data mechanism fail. An ablation traces most of an earlier accuracy loss on PEMS04 to a climatology input. Code, benchmark and forecasts: https://github.com/paradoxailab/paradox-world-model.

KEYWORDS

spatio-temporal forecasting, graph neural networks, time-series foundation models, benchmarking, probabilistic forecasting, conformal prediction, traffic, rainfall

1 INTRODUCTION

Forecasting how traffic or rainfall evolves over a physical network is a long-standing data-mining task with two competing recipes. Specialists, mostly graph neural networks and transformers, are trained on one network's history [Li et al. 2018; Liu et al. 2023a; Shao et al. 2022; Wu et al. 2019]. Time-series foundation models such as TimesFM [Das et al. 2024] are pretrained once on large corpora and forecast any series zero-shot. Each recipe is usually evaluated in its own protocol: specialists on point error over fixed windows, often with numbers copied from earlier papers, and foundation models on aggregate leaderboards of unrelated series. A practitioner choosing between them for a given network has no like-for-like evidence, and benchmark studies have shown that re-running published specialists under one pipeline changes their rankings [Shao et al. 2024].

We ask a narrow question with a fixed protocol: on real sensor networks, how does a small model trained from scratch on the network's own history compare with (a) published specialists run with their authors' code and (b) zero-shot foundation models, on the same windows, with the same point and probabilistic scores, when 30% of the sensors go dark, and with tests that say whether the differences are larger than noise?

Contributions.

- **PWM-Dyn** (§3), a 0.3–0.4M-parameter graph-diffusion model with explicit missing-data inputs, a three-quantile head and split-conformal calibration, trained in 9–52 GPU-minutes per seed on a free T4.
- **ParadoxBench** (§4), an open benchmark of six traffic networks and two monsoon rainfall regions with four tracks (standard context, open context, dead sensors, zero-shot transfer), one scorer for point, quantile and interval quality, saved forecasts and paired significance tests.
- **A controlled comparison** (§5) against STID, Graph WaveNet and STAEformer trained with their official code and published hyper-parameters on our windows, and against TimesFM 2.5 and 3.0 zero-shot. Every PWM-Dyn configuration choice was made on validation data and the reported models use seeds never used during development; four of the six traffic networks were never touched during development.
- **Findings, including losses.** Under a 12-step context a three-seed PWM-Dyn ensemble has the lowest MAE of the four trained models on five of six traffic networks and is second on PEMS08, where STAEformer leads by 1.4% even though its run was cut short; against STID, the closest rival, the margins are significant on three networks and not significant on PEMS08, PEMS04 and PEMS07, and single PWM-Dyn seeds are slightly worse than single STID runs on three networks. Against zero-shot TimesFM 3.0 it has lower MAE on seven of eight tasks (rainfall entries from the previous version) and lower interval score on seven of eight, and loses on PEMS03. With 30% of sensors dark it beats per-sensor climatology on seven of eight tasks while specialists without a missing-data mechanism fail, but climatology beats it on PEMS07. An ablation shows that a climatology input added in an earlier

version cost accuracy under the short context; removing it, chosen on validation, gave the reported model.

2 RELATED WORK

Spatio-temporal specialists. Traffic forecasting on sensor graphs moved from diffusion-convolutional recurrence (DCRNN [Li et al. 2018]) and graph convolution with temporal gating (STGCN [Yu et al. 2018]) to learned adjacency (Graph WaveNet [Wu et al. 2019], MTGNN [Wu et al. 2020], AGCRN [Bai et al. 2020]), attention (GMAN [Zheng et al. 2020], PDFormer [Jiang et al. 2023]) and, recently, simple identity-embedding models (STID [Shao et al. 2022]) and an embedding-heavy vanilla transformer (STAEformer [Liu et al. 2023a]). Benchmark studies report that gains between recent models are small and sensitive to protocol details, and that re-running models under one pipeline changes rankings [Liu et al. 2023b; Shao et al. 2024]. We follow that lesson: every opponent is run with its authors' code on our windows rather than quoted. Missing and dead sensors are usually treated as a separate kriging problem [Wu et al. 2021]; our robustness track scores it inside the forecasting task.

Time-series foundation models. TimesFM [Das et al. 2024; Google Research 2026], Chronos [Ansari et al. 2024], Moirai [Woo et al. 2024], Time-MoE [Shi et al. 2025] and Sundial [Liu et al. 2025] are pretrained on large corpora and forecast new series zero-shot, with long contexts and, in TimesFM 3.0, cross-series attention. They are rarely compared with trained graph specialists under identical windows, metrics and uncertainty scores; this paper does that for two TimesFM generations.

Uncertainty. Quantile regression [Koenker and Bassett 1978] with conformal calibration [Romano et al. 2019] gives finite-sample coverage under exchangeability; proper scores such as the pinball loss and the interval score [Gneiting and Raftery 2007] reward sharp, calibrated intervals. Traffic benchmarks almost always report point metrics only.

3 PWM-DYN

Problem. A network has N locations (road sensors or grid cells) joined by a fixed physical graph $A \in \mathbb{R}^{N\times N}$ built from road distance or geographic adjacency. From the last L observations $X \in \mathbb{R}^{L\times N}$, an observed-mask $O \in \{0,1\}^{L\times N}$ (0 = no reading), the position p of the last step in its daily or yearly cycle and the day of week w, the model returns three quantiles for each of the next H steps and every location,

$$\hat{Q} = f_\theta(X, O, p, w, A, C) \in \mathbb{R}^{H\times N\times 3}, \qquad \hat{Q}_{h,n,1} \le \hat{Q}_{h,n,2} \le \hat{Q}_{h,n,3}, \tag{1}$$

the 10th, 50th and 90th percentiles. $C \in \mathbb{R}^{H\times N}$ is the training-split climatology of the forecast steps: the per-location median at the same time of week (traffic, ±5 minutes) or day of year (rainfall, ±7 days), computed once from the training split. The median is the point forecast and $[\hat{Q}_{\cdot,\cdot,1}, \hat{Q}_{\cdot,\cdot,3}]$ the 80% prediction interval.

Architecture. PWM-Dyn (Figure 1) gives every location a d=96-dimensional state and mixes it with its physical neighbours. (i) A two-layer MLP reads the location's history concatenated with its mask, $[X \odot O;\ O]$, so a missing reading is an explicit zero flag rather than an imputed value; four scale-free cues (last value, masked mean, trend, observed share) enter through a separate linear map. (ii) Learned embeddings of p and w are added to every location. (iii) Optionally, a linear map of C is added. (iv) In the transductive variant, a per-location embedding and a learned low-rank graph $G = \mathrm{softmax}(\mathrm{relu}(SD^\top))$, $S, D \in \mathbb{R}^{N\times 16}$, are added; the inductive variant drops both so its weights run on unseen networks. (v) Three GraphMix blocks apply two-hop diffusion over A (and G), $h \leftarrow \mathrm{LN}(h + W[h; Ah; A^2h; Gh; G^2h])$, followed by a residual feed-forward layer. (vi) A linear head emits $H\times 3$ offsets from a base value and the outputs are sorted. The base is the last observed value; when the climatology input is used and a location's last reading is missing, the base is C, so a dark sensor is forecast as a deviation from its own normal pattern rather than from zero. Whether to use C is a configuration choice made on validation (§5): the reported standard-context model omits it and the long-context variant keeps it. The transductive model has 3.2×10^5 parameters on PEMS08.

Training. From random initialisation, with the pinball loss on the three quantiles plus an L_1 term on the median, over valid targets only (missing and zero readings excluded on traffic, as in the literature). During training each batch hides a random share $r \sim U(0, 0.3)$ of locations for the whole input window, which forces the model to reconstruct them from their neighbours. AdamW (weight decay 10^{-4}), one-cycle schedule with peak learning rate 2×10^{-3}, batch 32, gradient clipping at 5, a fixed budget of ≈33.5k optimisation steps (100 epochs on PEMS08); the checkpoint with the lowest validation MAE is kept.

Long-context variant. For the open-context track the input also carries the 24 values around the same time one day earlier (traffic) or the same week one year earlier (rainfall); total look-back 300 steps or 365 days, inside the foundation models' budget. Whether a task uses it is decided on validation.

Calibration. Quantile losses alone under-cover on heavy-tailed targets. We apply split-conformal quantile regression [Romano et al. 2019] on the validation split: per horizon, the score $s = \max(\hat{Q}_1 - y,\ y - \hat{Q}_3)$ in the model's target space gives an offset δ_h (its $\lceil (n+1)0.8 \rceil / n$ quantile), and the test interval becomes $[\hat{Q}_1 - \delta_h, \hat{Q}_3 + \delta_h]$. The median, hence MAE and RMSE, is unchanged. Each reported PWM-Dyn entry is the quantile mean of three independently trained seeds.

4 THE PARADOXBENCH BENCHMARK

ParadoxBench scores every model, specialist or foundation model, with one function on frozen windows. Version 1.1, used here, has eight tasks (Table 1): six traffic networks with their standard literature protocols (12 five-minute steps in, 12 out; chronological 60/20/20 split for PeMS flow, 70/10/20 for the two speed datasets) and two Indian monsoon rainfall regions from the IMD 0.25° daily grid [Nandi et al. 2024] (30 days in, 7 out; 2000–2018 train, 2019–2021 validation, 2022–2024 test). Values and time features of the four tasks added in 1.1 come from public copies of the datasets and road graphs from the DCRNN and ASTGNN releases [Guo et al. 2019; Li et al. 2018]; a script rebuilds them, so nothing is re-hosted.

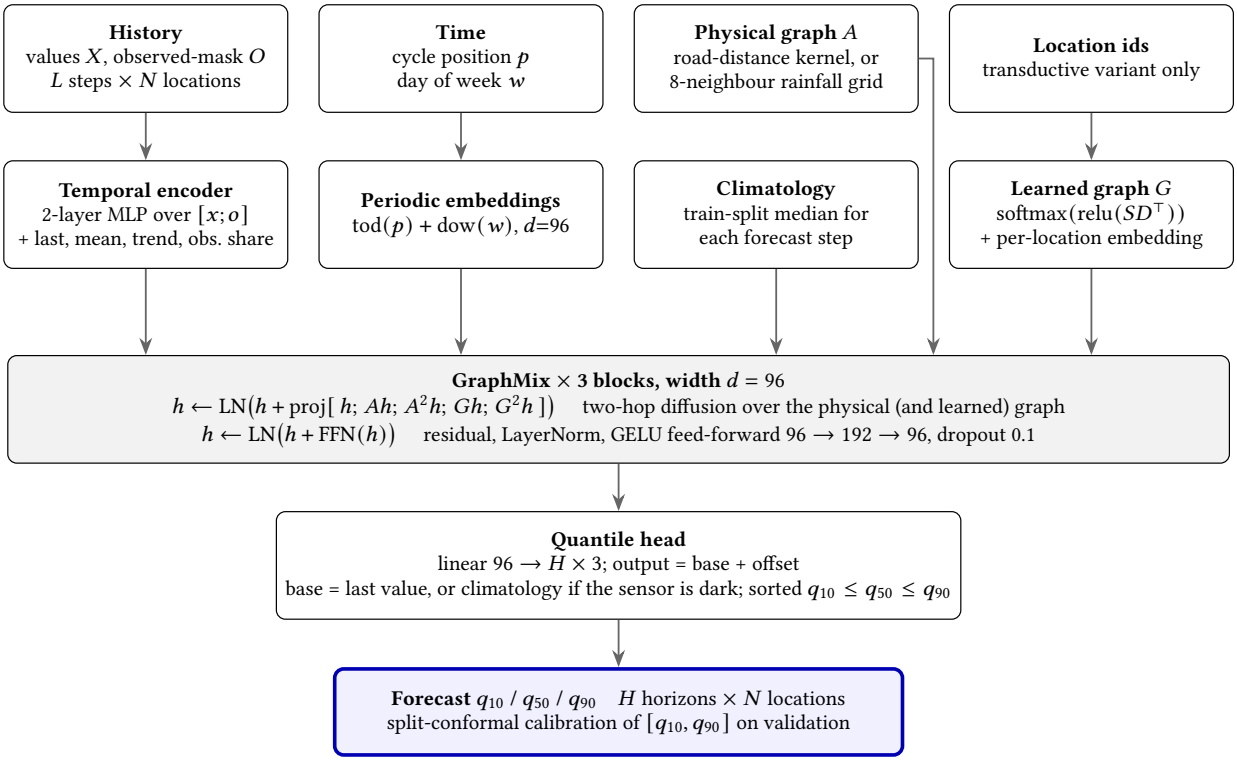


Figure 1: PWM-Dyn. Each location's history and observed-mask, the time, the training-split climatology and (in the transductive variant) a location embedding form a state that three GraphMix blocks diffuse over the physical graph A and a learned graph G; a linear head returns three quantiles per horizon as offsets from the last reading, or from climatology when the sensor is dark.

Table 1: ParadoxBench-1.1 tasks. Windows: official test windows (every 2nd; every 6th on the four tasks added in 1.1, so that foundation models fit a free-GPU budget). All models on a task are scored on the same windows; on METR-LA, 88 of them contain no valid (non-zero) target and drop out of every metric and test.

Task	Quantity	N	Steps	Windows
PEMS08 / PEMS04	flow, veh/5 min	170 / 307	17,856 / 16,992	1,775 / 1,688
PEMS03 / PEMS07	flow, veh/5 min	358 / 883	26,208 / 28,224	870 / 937
METR-LA / PEMS-BAY	speed, mph	207 / 325	34,272 / 52,116	1,139 / 1,734
IMD-CI / IMD-WC	rain, mm/day	land cells	9,132 days	530 / 530

Tracks. **A (standard)**: look-back limited to the standard context (12 steps, 30 days); any training on the task's training split. **B (open context)**: look-back up to 512 steps or 365 days; foundation models run zero-shot here. **R (dead sensors)**: 30% of locations, fixed by a seeded permutation, have no observations during the test period and are scored on those locations only. **Z (transfer)**: forecast PEMS08 with a model that has seen no PEMS08 data, not even its normalisation statistics or validation split.

Metrics. MAE, RMSE and MAPE of the median; QL, the mean pinball loss over the three quantiles (a proper score; 2 QL approximates CRPS); IS80, the interval score [Gneiting and Raftery 2007] of the 80% interval (width plus 10× the size of any miss); and COV80, the share of truths inside it. MAE is the headline point metric and IS80 the headline uncertainty metric; point-only models report MAE, RMSE and MAPE.

Protocol. Test data may not be used for training, selection, early stopping or calibration; validation data may. Each scored entry keeps its forecast file, so any two entries can be compared on the same windows. Paired comparisons use the Diebold–Mariano test [Diebold and Mariano 1995] on per-window absolute-error differences with a Newey–West [Newey and West 1987] long-run variance (one-day bandwidth), and a moving-block bootstrap [Künsch 1989] over whole days (2,000 resamples) for a 95% interval on the relative MAE difference.

5 EXPERIMENTS

Models. Specialists: STID [Shao et al. 2022], Graph WaveNet [Wu et al. 2019] and STAEformer [Liu et al. 2023a], imported unchanged from their official repositories (pinned commits in the supplement) and trained on each task's training split with the published hyperparameters, loss, optimiser, schedule and early stopping, on exactly the windows PWM-Dyn uses. The one code change is a compatibility fix in Graph WaveNet, whose 2-D-kernel `Conv1d` layers are rejected by current PyTorch and are built as `Conv2d`, the operation they computed. Each run had a wall-clock budget on one T4 (Table 6); runs that hit it keep their best-validation checkpoint and are marked. *Foundation models*: TimesFM 2.5 (200M) and TimesFM 3.0

in its multivariate mode, zero-shot, with a 512-step (365-day) context, run in a separate software environment. *References*: last value, seasonal naive and per-location climatology quantiles from the training split. *PWM-Dyn*: the configuration frozen by the ablation below, three fresh seeds (100–102), trained for the optimisation-step budget it has on PEMS08 (≈33.5k steps; 9–52 GPU-minutes per seed), conformally calibrated on validation and scored once.

5.1 Point accuracy

Table 2 gives test MAE and RMSE. Under the standard 12-step context (upper block) the PWM-Dyn ensemble has the lowest MAE on PEMS04, PEMS03, PEMS07, METR-LA and PEMS-BAY and is second on PEMS08 (14.12) behind STAEformer (13.93), which reached only 25 of its scheduled epochs; STAEformer's published PEMS08 result, 13.46 on all test windows, is better still. On the four networks never used during development the margin over the best trained opponent is 2.5% (PEMS03, Graph WaveNet), 0.3% (PEMS07, STID), 4.2% (METR-LA, Graph WaveNet) and 2.7% (PEMS-BAY, STID). STID, which completed its authors' schedule on four networks, is the closest rival on PEMS08, PEMS04, PEMS07 and PEMS-BAY and is within 0.4% of PWM-Dyn on PEMS08, PEMS04 and PEMS07, differences that are not significant (Table 3). Part of PWM-Dyn's margin is ensembling: its three single seeds average 14.33, 18.62 and 19.88 on PEMS08, PEMS04 and PEMS07, slightly worse than single-seed STID (14.18, 18.49, 19.62), while the ensemble costs about ten times STID's training time. In the open-context block, TimesFM 3.0 run zero-shot beats every trained model on PEMS03 (14.15) and is ahead of or within 4% of PWM-Dyn on four of six traffic networks; PWM-Dyn's long-context variant was preferred on validation only for PEMS08. RMSE orders the models similarly, with exceptions: all three specialists on PEMS04, and STID and STAEformer on PEMS08, have lower RMSE than PWM-Dyn, i.e. smaller errors on the largest deviations.

5.2 Is the difference larger than noise?

Table 3 tests each comparison on the per-window error differences. We call a difference significant only when the Diebold–Mariano test rejects at $p < 0.05$ *and* the day-block bootstrap interval excludes zero. Against STID, PWM-Dyn's advantage is significant on METR-LA, PEMS-BAY and PEMS03 ($p \leq 0.023$; on PEMS03 the interval is wide because the test period contains Thanksgiving week). On PEMS07 it is not (−0.3%, $p = 0.03$, interval $[-0.5, +0.0]$), and on PEMS08 and PEMS04 the two models are statistically tied ($p = 0.12$ and 0.82). STAEformer's lead on PEMS08 is significant. Against TimesFM 3.0 the advantage is significant on PEMS04, PEMS07, METR-LA and PEMS-BAY; on PEMS08 the interval excludes zero but $p = 0.06$; on PEMS03 TimesFM is ahead by 1.3%, not significantly. The standard deviation of single-seed test MAE is 0.002–0.22 (Table 5, right), small next to the margins over Graph WaveNet, STAEformer and TimesFM but as large as the gaps to STID on PEMS08, PEMS04 and PEMS07; the ensemble beats each of its seeds on every network.

5.3 Uncertainty and dead sensors

Intervals. After conformal calibration, PWM-Dyn's 80% intervals cover 78–85% of test values and have a lower interval score than TimesFM 3.0's on seven of eight tasks (Table 4; the two rainfall entries are from the previous model version); the exception is again PEMS03. TimesFM's intervals slightly over-cover (81–87%) everywhere; climatology intervals are wide and under-cover on traffic. Specialists give point forecasts only. **Dead sensors.** With 30% of locations dark throughout the test period, PWM-Dyn's MAE on those locations rises by 15–58% over its normal accuracy on traffic (and hardly at all on rainfall) but stays below per-sensor climatology on seven of eight tasks; on PEMS07 it is 6% worse than climatology. On PEMS08 and PEMS04 the previous version was also slightly better on dark sensors (18.70 and 24.99 against 19.25 and 25.33); our ablations do not isolate why. The standard-context model's lack of a climatology fallback is one candidate on PEMS04 and the new networks, but it does not explain PEMS08, where both versions keep it. The three specialists, fed zeros for the missing readings as their repositories do, produce errors of 5–310 (appendix): forecasting dark sensors needs an explicit mechanism, not just a graph.

5.4 What cost v0.4 its accuracy

An earlier version of PWM-Dyn added a per-location climatology input, and its test MAE under the 12-step context was 2–4% worse than the version before it. Table 5 separates the changes involved on validation. On PEMS04 the climatology input accounts for most of the loss: with it, validation MAE is 18.84 whichever checkpoint rule is used; without it, 18.39–18.61, with training length (60 vs 100 epochs) moving it by 0.22. On PEMS08 the effect is mixed: the climatology input improves validation MAE (14.45 vs 14.61) but not test MAE. The configuration with the lowest validation MAE averaged over both networks (no climatology input, 100 epochs, checkpoint chosen on every validation window) was frozen before any test scoring; its test MAE, reported for the record, is also the lowest of the four. The climatology input is not useless: it gives dark sensors a sensible baseline and remains in the long-context variant.

5.5 Cost

PWM-Dyn has 0.32–0.42M parameters, against 0.12–0.14M (STID), 0.31–0.32M (Graph WaveNet), 1.2–1.4M (STAEformer) and 200M (TimesFM 2.5). One seed trains in 9 (PEMS08), 9 (METR-LA), 13 (PEMS04), 14 (PEMS-BAY), 17 (PEMS03) and 52 (PEMS07) minutes on a T4, so the three-seed ensembles cost 0.4–2.6 GPU-hours per network; TimesFM 3.0 needed 0.6–2.0 GPU-hours per network for inference alone at our window counts. The study reported here ran in two free Kaggle sessions on 2×T4 (4.6 and 6.5 hours, about 22 T4-hours including TimesFM inference), plus the earlier runs that produced the rainfall entries.

6 LIMITATIONS

Compute-capped opponents. Each opponent run had a fixed wall-clock budget on one T4 (30–55 minutes for STAEformer and Graph WaveNet, 6–12 for STID; Table 6). STID reached its authors' 100-epoch schedule on four networks; Graph WaveNet on one; STAEformer on none, and it ran out of memory on PEMS07 at its published batch size. The comparison is therefore compute-matched rather than converged: STAEformer's numbers in particular understate what it reaches with its full schedule, and its published results

Table 2: Point accuracy: test MAE (RMSE below) on the official windows. Upper block: Track A, 12-step context, every model trained on the task's training split; lower block: Track B, foundation models zero-shot with 512 steps (365 days on rainfall). Bold: best MAE in the block. † stopped by its wall-clock budget before its authors' schedule ended (Table 6). * network never used during development. Rainfall PWM-Dyn entries are v0.4; rainfall has no specialist entries. Units: vehicles/5 min (PeMS flow), mph (METR-LA, PEMS-BAY), mm/day (IMD).

Model	PEMS08	PEMS04	PEMS03*	PEMS07*	METR-LA*	PEMS-BAY*	IMD-CI	IMD-WC
Last value	25.30	31.56	23.89	35.63	5.04	2.17	4.37	4.88
	38.66	48.25	37.31	54.37	10.98	5.11	13.55	14.94
Climatology	22.18	25.02	25.93	29.98	3.90	2.53	2.89	3.65
	40.77	41.36	47.80	58.67	8.41	5.48	10.70	13.00
STID	14.18	18.49	15.50	19.62†	3.12	1.56†	–	–
	23.21	30.05	26.83	32.72	6.50	3.60		
Graph WaveNet	14.77	19.37†	14.70†	23.39†	3.05†	1.57†	–	–
	23.59	30.75	25.17	36.92	6.15	3.57		
STAEformer	**13.93**†	19.21†	15.45†	–	3.09†	1.59†	–	–
	23.17	30.66	26.03		6.33	3.63		
PWM-Dyn (ours)	14.12	**18.46**	**14.34**	**19.56**	**2.92**	**1.52**	**2.84**	**3.41**
	23.46	30.77	23.12	32.60	5.88	3.45	10.42	12.01
TimesFM 2.5	15.04	19.70	–	–	–	–	2.96	3.67
	24.09	31.70					10.58	12.65
TimesFM 3.0 mv	14.55	19.13	**14.15**	20.27	3.40	1.66	2.92	3.62
	23.57	31.02	22.70	33.13	6.97	3.95	10.65	12.78
PWM-Dyn (ours), Track B	**14.18**	**18.46**	14.34	**19.56**	**2.92**	**1.52**	**2.84**	**3.41**
	23.38	30.77	23.12	32.60	5.88	3.45	10.42	12.01

Table 3: Paired tests of PWM-Dyn against each model on the same windows: relative MAE difference (negative = PWM-Dyn better) and Diebold–Mariano p-value; in brackets the 95% day-block bootstrap interval. Track A for specialists, Track B for TimesFM. † opponent budget-capped.

Task	STID	GWNet	STAEf.	TimesFM 3.0
PEMS08	-0.4%, $p = 0.12$ [-1.0, +0.1]	-4.4%, $p < 10^{-3}$ [-4.9, -4.0]	$+1.4\%$, $p < 10^{-3\dagger}$ [+0.7, +1.9]	-2.5%, $p = 0.06$ [-5.4, -0.3]
PEMS04	-0.2%, $p = 0.82$ [-2.0, +1.1]	-4.7%, $p < 10^{-3\dagger}$ [-5.3, -4.3]	-3.9%, $p < 10^{-3\dagger}$ [-4.7, -3.5]	-3.5%, $p = 0.03$ [-7.0, -0.8]
PEMS03*	-7.5%, $p = 0.02$ [-14.1, -2.3]	-2.5%, $p < 10^{-3\dagger}$ [-3.3, -1.2]	-7.2%, $p < 10^{-3\dagger}$ [-10.8, -4.6]	$+1.3\%$, $p = 0.48$ [-1.8, +5.3]
PEMS07*	-0.3%, $p = 0.03^{\dagger}$ [-0.5, +0.0]	-16.4%, $p < 10^{-3\dagger}$ [-17.5, -15.4]	–	-3.5%, $p < 10^{-3}$ [-5.4, -1.6]
METR-LA*	-6.6%, $p < 10^{-3}$ [-7.6, -5.5]	-4.2%, $p < 10^{-3\dagger}$ [-5.0, -3.3]	-5.5%, $p < 10^{-3\dagger}$ [-6.2, -4.8]	-14.2%, $p < 10^{-3}$ [-16.3, -11.5]
PEMS-BAY*	-2.7%, $p < 10^{-3\dagger}$ [-3.6, -1.9]	-3.2%, $p < 10^{-3\dagger}$ [-3.9, -2.2]	-4.7%, $p < 10^{-3\dagger}$ [-5.2, -4.2]	-8.5%, $p < 10^{-3}$ [-11.4, -5.2]

Table 4: Probabilistic quality (Track B, IS80 with 80% coverage) and dead sensors (Track R, MAE on the 30% of locations with no test-period readings). Specialists without a missing-data mechanism are fed zeros, their repositories' missing-value convention.

	IS80 (coverage %)			Track R MAE	
Task	PWM-Dyn	TimesFM 3.0	Climat.	PWM-Dyn	best other
PEMS08	66.94 (80)	67.84 (81)	117.83 (71)	**19.25**	21.78 (clim.)
PEMS04	86.23 (80)	88.42 (81)	127.81 (72)	**25.33**	27.39 (clim.)
PEMS03*	67.14 (78)	65.50 (81)	154.10 (61)	**17.71**	23.31 (clim.)
PEMS07*	93.99 (80)	95.32 (81)	159.70 (76)	30.95	**29.10** (clim.)
METR-LA*	14.34 (81)	17.37 (81)	19.42 (75)	**3.36**	4.07 (clim.)
PEMS-BAY*	7.54 (79)	8.29 (81)	13.19 (75)	**2.31**	2.71 (clim.)
IMD-CI	18.01 (85)	19.79 (86)	18.21 (91)	**2.84**	2.88 (clim.)
IMD-WC	20.27 (83)	21.85 (87)	21.09 (91)	**3.43**	3.60 (clim.)

(e.g. 13.46 MAE on PEMS08 with all test windows) remain ahead of ours where reported. Each opponent was run with one seed.

Development exposure. PEMS08 and PEMS04 shaped PWM-Dyn's design in earlier versions, including one look at PEMS08 test scores of development checkpoints; the reported models use fresh seeds and a configuration chosen on validation, but the design itself is not blind on those two networks. PEMS03, PEMS07, METR-LA and PEMS-BAY were never used during development.

Scope. Traffic tasks use the 12-step literature protocol and the new tasks score every sixth window. Rainfall covers two Indian regions and its PWM-Dyn entries come from the previous model version. TimesFM 3.0 is used zero-shot under a non-commercial research licence; we did not fine-tune it, and fine-tuned foundation models may close the gap we report. A PWM-Dyn win over a foundation model is a specialist-versus-zero-shot result, not evidence of better generalisation; the transfer track shows the opposite.

7 CONCLUSION

Trained on a network's own history and scored on identical windows, a 0.3–0.4M-parameter graph model with explicit missing-data inputs and conformal intervals is, as a three-seed ensemble, the

Table 5: Left: what cost v0.4 accuracy. Validation MAE (selection criterion, seeds 0–1) and test MAE of each configuration's ensemble; the frozen choice is marked. Right: test MAE of the three fresh seeds of the frozen model and of their ensemble (Track A). The v0.4.1 row is its own three-seed run (seeds 0–2).

Configuration	Val. MAE		Test MAE	
	P08	P04	P08	P04
clim., 100 ep., select every 4th (v0.4.1)	14.52	18.84	14.50	19.02
clim., 100 ep., select every window	14.45	18.84	14.45	19.06
no clim., 60 ep., select every window	14.60	18.61	14.23	18.47
no clim., 100 ep., select every window (frozen)	14.61	18.39	14.14	18.42

Task	Seeds (test MAE)	Mean ± SD	Ens.
PEMS08	14.22 / 14.24 / 14.52	14.33 ± 0.17	14.12
PEMS04	18.52 / 18.70 / 18.65	18.62 ± 0.09	18.46
PEMS03*	14.45 / 14.45 / 14.65	14.52 ± 0.12	14.34
PEMS07*	20.12 / 19.69 / 19.83	19.88 ± 0.22	19.56
METR-LA*	2.960 / 2.959 / 2.956	2.958 ± 0.002	2.919
PEMS-BAY*	1.544 / 1.536 / 1.535	1.538 ± 0.005	1.519

most accurate of the trained models we ran on five of six traffic networks, is not significantly different from STID on PEMS08, PEMS04 and PEMS07, and beats a 2026 zero-shot foundation model on most tasks in both point and interval quality, while still forecasting dark sensors better than climatology on all but one task. It is not state of the art in absolute terms, loses to STAEformer on PEMS08 and to TimesFM 3.0 on PEMS03, owes part of its margin over STID to ensembling, and its advantage over foundation models depends on having the network's own history. The benchmark, its scorer, every opponent harness and all saved forecasts are released so that these comparisons can be re-run and extended.

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Benchmark opponents use the official code of STID, Graph WaveNet and STAEformer and the open weights of Google's TimesFM. Data: Caltrans PeMS; the METR-LA and PEMS-BAY releases of DCRNN; India Meteorological Department (via imdlib). Use of AI tools: Claude (Anthropic) assisted with writing code, running experiments and drafting the text. Every table is regenerated by a script from the logged result lines, and the authors take full responsibility for the content. Project page: https://paradoxailab.com.

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A OPPONENT TRAINING

Each opponent's model class is imported unchanged from a pinned commit of its official repository (STID `e8b313b`, Graph WaveNet `6b162e8`, STAEformer `fc49d39`). The training loop re-implements each repository's train script: STID uses Adam (lr 2×10^{-3}, weight decay 10^{-4}), MultiStepLR at epochs 1/50/80 ($\gamma = 0.5$), batch 64, masked MAE, clipping at 5, 100 epochs; Graph WaveNet uses Adam (lr 10^{-3}, weight decay 10^{-4}), batch 64, masked MAE, clipping at 5, 100 epochs, the doubly-stochastic transition supports of the road graph plus the adaptive adjacency; STAEformer uses the per-dataset settings of its configuration file (Adam, lr 10^{-3}, dataset-specific

Table 6: Opponent training: epochs completed and what stopped the run (A = authors' schedule, B = wall-clock budget, minutes in brackets). One seed each, official code and hyper-parameters, mixed precision.

Task	STID	Graph WaveNet	STAEformer
PEMS08	100 A (6)	100 A (30)	25 B (30)
PEMS04	100 A (6)	64 B (30)	12 B (30)
PEMS03*	100 A (8)	58 B (40)	10 B (40)
PEMS07*	84 B (12)	20 B (50)	out of memory
METR-LA*	100 A (8)	60 B (40)	12 B (40)
PEMS-BAY*	94 B (12)	30 B (50)	6 B (55)

weight decay and milestones, batch 16, Huber loss on PeMS and masked MAE on METR-LA/PEMS-BAY, early stopping with the published patience). All use mixed precision and keep the checkpoint with the lowest validation MAE. Inputs are the raw series z-scored with training-split statistics (zeros included, as in the repositories) plus time-of-day and day-of-week features. Table 6 lists epochs and stopping causes.

B DEAD SENSORS, ALL ENTRIES

Table 7: Track R (30% of locations dark during the test period): MAE on the dark locations. Specialists receive zeros for missing readings, their repositories' convention.

Task	PWM-Dyn	Climatology	STID	Graph WaveNet	STAEformer
PEMS08	19.25	21.78	198.09	216.21	186.51
PEMS04	25.33	27.39	120.58	132.51	188.00
PEMS03*	17.71	23.31	113.05	138.37	119.34
PEMS07*	30.95	29.10	293.03	310.03	–
METR-LA*	3.36	4.07	5.11	5.89	5.33
PEMS-BAY*	2.31	2.71	42.19	51.76	44.72

C REPRODUCIBILITY

The benchmark package, the PWM-Dyn code, the opponent harnesses, the analysis script that produced every table, and the saved forecasts of every entry are public at https://github.com/paradoxailab/paradox-world-model (folder `v0.6-conference`). A provenance check verifies that PWM-Dyn's code imports no external model library and that its checkpoints contain only its own parameters; it passes for every reported model. Data are rebuilt from public sources at run time. Hardware: Kaggle, 2× NVIDIA T4 per session; PWM-Dyn session 4.6 h, opponent session 6.5 h, analysis on CPU.